

FAQ-10: How Do I Size and Scale ActiveGates?

Series: FAQ — Frequently Asked Questions | **Reference:** 10 — ActiveGate Sizing and Scaling |
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Overview

"How big should my ActiveGate be?" is really three questions wearing one coat: **what jobs** will this ActiveGate do, **how much volume** does each job carry, and **how much headroom** do you need for failover and growth. Dynatrace publishes baseline sizing for each job — but the numbers live on different documentation pages (host-based routing, containerized Kubernetes monitoring, private synthetic locations), and none of them tells you when an ActiveGate you already run has outgrown its size.

This entry consolidates the decision: which dimension drives sizing for each ActiveGate capability, the current documented baselines for host-based, containerized, and synthetic-enabled ActiveGates, how high availability and network zones change the math, and — the part teams most often skip — the `dt.sfm.active_gate.*` self-monitoring metrics that tell you an ActiveGate is saturating *before* data starts dropping.

Deployment mechanics stay with their owning series: ONBRD-03 (installing and deploying ActiveGates), the K8S series (DynaKube and containerized ActiveGates), SYNTH (private synthetic locations), and FAQ-05 (managing ActiveGate updates). This entry owns the sizing decision itself.

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Prerequisites

Requirement	Details
Audience	Platform engineers, Dynatrace administrators, and architects planning or operating ActiveGate fleets
Format	Decision-support document — consolidates sizing baselines and scaling signals; not a hands-on installation lab
Deployment	Dynatrace SaaS; Environment ActiveGates (host-based and/or containerized)
Related topic series	ONBRD-03 (deploying ActiveGate — installation, groups, example deployment), K8S (DynaKube and containerized ActiveGate mechanics), SYNTH-04 (private synthetic locations), M2S-03 / S2S-04 (ActiveGate design during migration)
Related FAQ	FAQ-05 (managing ActiveGate updates on SaaS — roles, sequencing, update rings)

1. Short Answer

Size an ActiveGate in three steps:

1. **Decide what the ActiveGate will do.** Each capability has its own sizing dimension — you cannot size an ActiveGate without knowing its job list, and demanding capabilities (synthetic, Kubernetes platform monitoring) deserve dedicated ActiveGates rather than sharing one box.
2. **Start from the documented baseline for that dimension.** Hosts routed → the host-based tables (§3). Pods monitored → the Kubernetes S/M/L scenarios (§4). Browser monitors per hour → the synthetic node sizes (§5).
3. **Provision headroom and verify with self-monitoring.** Dynatrace's guidance is that the ActiveGate machine *"should not exceed 50% CPU and 80% memory"* — treat those as your alert thresholds, watch the self-monitoring metrics (§7), and scale out for routing or up for Kubernetes monitoring when you approach them (§8).

The capability-to-dimension map:

ActiveGate job	What drives its sizing	Baseline
Routing OneAgent traffic	Number of connected OneAgents (hosts) and their data volume	§3
Kubernetes monitoring	Pod count (primary), node count (secondary), Prometheus-annotated pods	§4

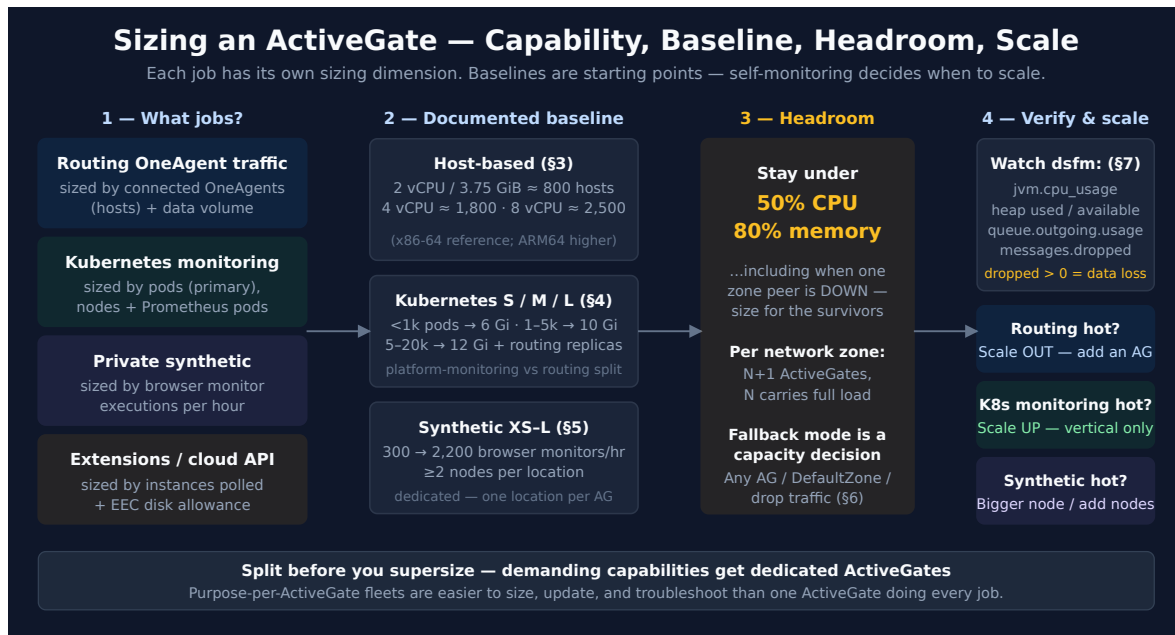
ActiveGate job	What drives its sizing	Baseline
Cloud & remote API monitoring (AWS, Azure, VMware, SNMP, Prometheus...)	Number of monitored endpoints/entities polled per cycle	§3 + extension notes — see the 1.345 module note in §2
Extensions (EEC)	Extension instances and monitored devices; adds disk requirements	§3
Private synthetic	Browser monitors per hour (HTTP monitors are far cheaper)	§5
z/OS routing	Mainframe traffic via the zRemote module	Size with Dynatrace guidance for your mainframe volume

One rule sits above all the tables: **sizing numbers are starting points, not guarantees**. Actual capacity depends on your data volume per host, extension mix, and network — which is why §7 (measuring the real utilization) is part of sizing, not an afterthought.

Scope — Dynatrace SaaS. This entry covers **Environment ActiveGates on SaaS**. On Dynatrace Managed the Environment-ActiveGate baselines broadly transfer (same machines, same jobs), but Managed adds two sizing domains this entry deliberately doesn't cover — **Cluster ActiveGates** (the on-premise clustering component) and the sizing of the Managed cluster nodes themselves — and §7's Grail-native `dt.sfm.*` metrics aren't available there, leaving the classic `dsfm:` family as the self-monitoring surface. If you're consolidating from Managed to SaaS, the M2S series owns that journey.

Sources: [Linux ActiveGate hardware and system requirements \(DT docs\)](#), [Dynatrace ActiveGate \(DT docs\)](#) — incl. the Environment vs Cluster ActiveGate split. **Derived:** the three-step framing and the capability-to-dimension map synthesize the per-capability sizing pages cited throughout this entry; the Managed scope note combines the hub's ActiveGate-type split with §7's Grail-bucket residency (Managed environments don't include Grail).

2. What Drives ActiveGate Load



The ActiveGate hub enumerates its jobs: routing OneAgent traffic as a secure proxy, monitoring cloud environments and remote technologies via API (AWS, VMware, Azure, Kubernetes, OpenShift, Google Cloud, SNMP, Prometheus, and more), running synthetic monitors from private locations, routing z/OS traffic, and providing Dynatrace API access. Each job consumes a different resource profile:

Classic-only modules on Latest Dynatrace — breaking (ActiveGate 1.345): ActiveGate 1.345 disallows modules that exclusively serve Dynatrace Classic monitoring pages on ActiveGates connected to a *Latest Dynatrace* environment — **AWS, Azure, Cloud Foundry, VMware, memory dumps, and database insights**. Disable or uninstall those modules *before* upgrading such an ActiveGate; the upgrade does not complete cleanly while they remain enabled. ActiveGate 1.345 released 08/12/2026 with a **staged rollout from 08/25/2026** — check which version your ActiveGates are actually on before planning the work, and note this changes *what an ActiveGate may run*, not the sizing math below. Classic-operated environments are unaffected. The release note does not name replacement modules for AWS/Azure/VMware on Latest Dynatrace; confirm the supported ingest path for those technologies against the ActiveGate capabilities page before you remove a working module.

- **Routing** is network- and CPU-bound: load scales with the number of OneAgents connected and the volume of data they send. Memory matters for the outgoing message queue when the uplink is slow.
- **Kubernetes monitoring** is memory- and CPU-bound on the *monitored workload count*: Dynatrace's Kubernetes sizing guide is explicit that consumption "*scales with the number of pods primarily due to increased data processing and storage needs,*" with node count a secondary driver and Prometheus metric collection adding CPU as annotated-pod counts rise.
- **Extensions (EEC)** add per-instance polling work plus disk: the Extension Execution Controller wants ~2.1 GB of free disk for its persistence mechanism, and the extensions module claims its own

installation and log/cache space.

- **Synthetic** is the most demanding per unit of work — real browsers rendering real pages. Dynatrace states plainly that *"a Synthetic-enabled ActiveGate has more demanding hardware and system requirements than a regular Environment or Cluster ActiveGate."*

Two placement consequences follow. First, Dynatrace recommends a **dedicated system**: running ActiveGate alone on its machine is both a performance and a security recommendation. Second, **don't stack demanding capabilities on one ActiveGate** — the Kubernetes guide's recommended production pattern is itself a split (one ActiveGate for platform monitoring, separate replicas for OneAgent routing, §4), and synthetic locations can't share an ActiveGate with other locations at all (§5). A routing-only ActiveGate plus purpose-built ActiveGates for heavy jobs is easier to size, easier to diagnose, and safer to update (FAQ-05's update-ring guidance assumes the same separation).

Sources: [Dynatrace ActiveGate \(DT docs\)](#), [Sizing guide for ActiveGates in the Kubernetes monitoring use-case \(DT docs\)](#), [Requirements for private Synthetic locations \(DT docs\)](#), [Linux ActiveGate hardware and system requirements \(DT docs\)](#), [What's new in ActiveGate 1.345 \(DT docs\)](#) — classic-only modules disallowed on Latest Dynatrace.

Derived: the per-capability resource-profile characterizations combine the cited sizing drivers with the deployment-separation recommendations.

3. Baseline Sizing — Host-Based ActiveGates

Minimums (routing/monitoring ActiveGate): 2 GB RAM (4 GB recommended), one dual-core processor, 4 GB free disk for installation/configuration/logs plus 4 GB for cached OneAgent installers and container images if the ActiveGate serves them.

Capacity by machine size. Dynatrace publishes estimated host counts per reference instance size, and the machine *"should not exceed 50% CPU and 80% memory"* in steady state:

Architecture	Reference instance	vCPU	RAM (GiB)	Estimated hosts routed
x86-64	c6i.large	2	3.75	~800
x86-64	c6i.xlarge	4	7.5	~1,800
x86-64	c6i.2xlarge	8	15	~2,500
ARM64	c7g.large	2	3.75	~1,300
ARM64	c7g.xlarge	4	7.5	~2,700
ARM64	c7g.2xlarge	8	15	~5,500
s390	S	2	4	~800
s390	M	4	8	~1,500

Reading the table honestly: the reference points are AWS instance shapes, so for on-premises VMs map by vCPU/RAM, and treat "estimated hosts" as exactly that — an estimate at typical per-host data volume. Hosts running chatty workloads (heavy log routing through the ActiveGate, dense process instrumentation) consume capacity faster.

Disk beyond the minimums. The requirements page itemizes where space goes — installation (~600 MB executables), logs (1.2 GB), auto-update downloads (600 MB), temporary files (4 GB) — and Environment ActiveGates running extensions add ~1.2 GB for the extensions module plus 2 GB for its logs/cache. If extension-data persistence is enabled, the EEC wants a further *"2136 MB of free disk space: 600 MB for reliability mechanism, 1.5 GB as buffer"* — without it, persistence disables itself. A practical floor for an extensions-running ActiveGate is ~15 GB of free disk; disk is cheap, resizing later is not.

Windows ActiveGates follow the same sizing model with their own OS-support matrix — see the Windows requirements page alongside the Linux one.

Update (SaaS 1.343, July 2026): Dynatrace published official **sizing guides for Environment ActiveGates** with capacity recommendations specifically for **log ingestion** infrastructure, plus an upgraded **log ingest monitoring dashboard** covering OneAgent modules and Environment ActiveGate. If your AG fleet fronts log ingestion, size against those guides first — they postdate (and refine) the general host-based baselines in this section for that workload.

Sources: [Linux ActiveGate hardware and system requirements \(DT docs\)](#) — sizing tables, 50%/80% guideline, disk itemization, and EEC persistence figures quoted from this page. **Derived:** the ~15 GB practical disk floor sums the itemized directories with growth margin; the "chatty workloads consume capacity faster" caveat generalizes the per-host-volume dependency the docs imply but don't quantify.

4. Sizing Containerized ActiveGates on Kubernetes

Containerized ActiveGates (deployed via DynaKube — mechanics in the K8S series) size against the **monitored cluster's workload count**, not host counts. Dynatrace's Kubernetes sizing guide sets three scenarios:

Scenario	Pods monitored	CPU request	CPU limit	Memory request = limit
S	< 1,000	200m	1000m	6 Gi
M	1,000–5,000	1000m	2000m	10 Gi
L	5,000–20,000	2000m	4000m	12 Gi

Operational notes straight from the guide:

- **Prefer no CPU limits** where cluster policy allows — throttling an ActiveGate delays its processing pipeline.

- **One ActiveGate per cluster.** A single ActiveGate *can* monitor multiple clusters, but the guide marks that pattern **not recommended**.
- **Split platform monitoring from routing** for production application-observability clusters: a Kubernetes-monitoring ActiveGate (scales **vertically only** — the table above) plus a separate routing ActiveGate deployment for OneAgent traffic that scales **horizontally**:

Pods	Routing CPU request/limit	Memory request = limit	Replicas
< 1,000	250m / 1000m	2 Gi	3
1,000–5,000	500m / 2000m	4 Gi	3
5,000–20,000	1000m / 4000m	6 Gi	6

The guide also publishes its own health thresholds — effectively the containerized version of §7: raise the CPU request when usage exceeds **85%** of it or CPU throttling exceeds **10%**; raise the memory request when the working set passes **80%**; and treat a Kubernetes-monitoring pipeline execution time beyond **50–60 seconds** as a CPU-shortage signal.

Rolling out (ActiveGate 1.343) — re-baseline the percentage thresholds, not the absolute figures. ActiveGate 1.343 (published 07/15/2026, rollout from 07/28/2026) changes the Kubernetes CPU and memory **request and limit** metrics to **include init containers**. Reported reservations therefore rise, and because the health thresholds above are expressed as a *percentage of the request*, the same absolute usage now reads as a **lower** percentage — the tripwires effectively move without anyone changing them.

Three things to hold separate:

- **The scenario tables above are unaffected.** They are absolute CPU/memory figures, not ratios, so the S/M/L values and the routing replica counts stand as published.
- **The `*_usage` and `*_working_set` metrics are unaffected** — the change is to the request/limit side of the ratio, i.e. the denominator, not to consumption.
- **The percentage thresholds need re-baselining.** Take a fresh reading after the change reaches your clusters rather than carrying pre-1.343 percentages forward. A workload that sat at 84% of request before the change may report comfortably lower afterwards while consuming exactly as much as it did — and a threshold tuned against the old denominator will now fire later than you intended, which is the more dangerous direction.

Verify the ActiveGate version running in the cluster before deciding which side of the change a reading came from; until it lands, the thresholds above remain correct as written.

Sources: [Sizing guide for ActiveGates in the Kubernetes monitoring use-case \(DT docs\)](#) — scenario tables, replica counts, no-CPU-limit preference, single-cluster recommendation, and health thresholds are all from this page, [ActiveGate 1.343 release notes \(DT docs\)](#) — Kubernetes CPU/memory request and limit metrics now include init containers. **Derived:** the "re-baseline the percentages, the absolute tables stand" conclusion follows from the thresholds being ratios whose denominator the 1.343 change raises.

5. Synthetic-Enabled ActiveGates

Synthetic is the one capability where the sizing unit is **work per hour**, and where Dynatrace publishes discrete node sizes. All sizes below sustain up to ~300k HTTP monitor executions per hour — the browser-monitor column is what separates them, because each browser monitor drives a real browser:

Node size	vCPU	RAM	Free disk	Max browser monitors/hr
XS	2	4 GB	20 GB	300
S	4	8 GB	25 GB	650
M	8	16 GB	30 GB	1,200
L	16	32 GB	40 GB	2,200

Planning rules that change the architecture, not just the size:

- **Dedicate the ActiveGate.** Synthetic-enabled ActiveGates are *"more demanding"* than routing ActiveGates, support a narrower OS matrix, and *"you can't use one ActiveGate for multiple locations"* — so a synthetic node does synthetic, for exactly one location.
- **At least two nodes per location.** Dynatrace: *"We recommend using at least two ActiveGates for a location. Additional ActiveGates are used for failover and load balancing."* Size the location so the monitor schedule still fits if one node is down.
- **Budget by the browser column.** Convert your monitor plan to executions/hour (monitors × frequency × steps where relevant) and pick the node size with comfortable margin — then scale the *location* by adding nodes rather than pushing one node to its ceiling.

Deployment and location design live in SYNTH-04; containerized synthetic locations on Kubernetes have their own requirements page.

Sources: [Requirements for private Synthetic locations \(DT docs\)](#) — node-size table quoted; [Private Synthetic locations \(DT docs\)](#) — two-node recommendation and one-location-per-ActiveGate constraint quoted. **Derived:** the executions-per-hour budgeting recipe operationalizes the published per-node ceilings.

6. High Availability, Network Zones, and Placement

Sizing an individual ActiveGate is half the job; the other half is sizing the **group of ActiveGates a zone can fail over across**.

How failover works. Network zones *"prevent inefficient connections by defining the ActiveGate that each OneAgent should use."* When no ActiveGate in the primary zone is reachable, OneAgents try the zone's configured **alternative zones**, then behave per the zone's **fallback mode**: *Any ActiveGate* (default — traffic routes to any available ActiveGate), *Only DefaultZone*, or *None (drop traffic)*. Two sizing consequences:

- **Survivor capacity is the real requirement.** If a zone runs two ActiveGates at 45% CPU each and one dies, the survivor takes (roughly) the full load — past the 50% guideline and climbing. In community practice the working rule is **N+1 per zone, where N alone can carry the zone's whole load at ≤50% CPU** — the docs define the failover mechanics but don't publish a per-zone count; verify the headroom against your own self-monitoring numbers (§7).
- **Fallback mode is a capacity decision, not just a security one.** Any *ActiveGate* means a zone failure silently lands its load on ActiveGates you sized for other zones. *None (drop traffic)* protects the fleet but drops data. Pick deliberately, and if you rely on alternative zones, make sure those ActiveGates are both **reachable** from the primary zone's OneAgents (a documented requirement) and **sized** to absorb the addition.

Placement basics (depth in ONBRD-03 and M2S-03): one or more ActiveGates per network zone/datacenter close to the OneAgents they serve; ActiveGate **groups** to control which ActiveGates handle which purpose; and remember that a proxy or firewall between OneAgents and the ActiveGate shapes effective throughput as much as the machine size does.

Sources: [Get started with network zones \(DT docs\)](#) — failover behavior, alternative zones, fallback modes, and the reachability requirement; [Linux ActiveGate hardware and system requirements \(DT docs\)](#) — the 50% CPU guideline the survivor math builds on. **Derived:** the survivor-capacity rule and N+1 guidance are community practice composed from the documented failover model plus the headroom guideline.

7. Detecting Saturation — Self-Monitoring Signals

Every SaaS and Managed environment ships ActiveGate **self-monitoring metrics** — no setup required. They exist on two surfaces: a Grail-native **Gen3** family, `dt.sfm.active_gate.*`, and a classic **Gen2** family, `dsfm:active_gate.*` (covered at the end of this section). This entry works in the Gen3 namespace throughout — it is what Dynatrace's own diagnostic dashboard queries, it lives in Grail metric buckets (`dt_system_metrics` / `default_metrics`) where DQL `timeseries` reaches it directly, and every key carries fleet-slicing dimensions (`dt.active_gate.id` , `host.name` , `dt.active_gate.group.name` , `dt.network_zone.id`). Four signals, watched per ActiveGate, answer "is this ActiveGate the right size" continuously:

Metric	What it tells you	Act when
<code>dt.sfm.active_gate.jvm.cpu_usage</code>	ActiveGate Java process CPU	Sustained peaks above the 50% guideline
<code>dt.sfm.active_gate.jvm.heap_memory_used / ..._available</code>	Heap pressure	Used/available ratio trending past 80%

Metric	What it tells you	Act when
<code>dt.sfm.active_gate.communication.queue.outgoing.usage</code>	Messages queued faster than they're being sent out	Queue grows and doesn't drain
<code>dt.sfm.active_gate.communication.messages.dropped</code>	Messages dropped — data loss	Any non-zero value

`dt.sfm.active_gate.communication.agent_modules.connected` rounds out the picture by showing how OneAgent load is actually distributed across the fleet.

```
// ActiveGates whose JVM CPU peaked above the 50% headroom guideline (24h)
timeseries cpu = avg(dt.sfm.active_gate.jvm.cpu_usage), from:-24h, by:{dt.active_gate.id, host.name}
| fieldsAdd peak = arrayMax(cpu)
| filter peak > 50
| sort peak desc
```

```
// Heap utilization per ActiveGate — compare against the 80% memory guideline
timeseries { used = avg(dt.sfm.active_gate.jvm.heap_memory_used), available =
avg(dt.sfm.active_gate.jvm.heap_memory_available) }, from:-24h, by:{dt.active_gate.id, host.name}
| fieldsAdd heap_used_pct = round(arrayAvg(used) / arrayAvg(available) * 100, decimals: 1)
| fieldsRemove used, available
| sort heap_used_pct desc
```

```
// Backpressure check — growing outgoing queue or any dropped messages (7d)
// default: 0 because drop metrics are sparse — they only report when drops occur
timeseries { queued = avg(dt.sfm.active_gate.communication.queue.outgoing.usage), dropped =
sum(dt.sfm.active_gate.communication.messages.dropped, default: 0) }, from:-7d, by:
{dt.active_gate.id, host.name}, union:true
| fieldsAdd total_dropped = arraySum(dropped), avg_queue = round(arrayAvg(queued),
decimals: 0)
| filter total_dropped > 0 OR avg_queue > 0
| sort total_dropped desc
```

```
// How OneAgent load is distributed across the fleet, per zone (24h)
timeseries agents = avg(dt.sfm.active_gate.communication.agent_modules.connected),
from:-24h, by:{dt.active_gate.id, host.name, dt.network_zone.id}
| fieldsAdd avg_connected = round(arrayAvg(agents), decimals: 0), network_zone =
if(isNull(dt.network_zone.id), "default", else: dt.network_zone.id)
| fieldsRemove dt.network_zone.id
| sort avg_connected desc
```

Wire the first three into anomaly-detection or a scheduled workflow (the ALERT and WFLOW series own the routing patterns) so a saturating ActiveGate pages someone before `messages.dropped` goes non-

zero. For containerized ActiveGates, layer on the Kubernetes-native thresholds from §4 (CPU >85% of request, throttling >10%, working set >80%, pipeline >50–60 s).

The full catalog — and the ready-made diagnostic dashboard

The namespace goes far beyond the four core signals, and it is exactly what Dynatrace's own **"ActiveGate diagnostic overview" dashboard** is built on (the dashboard ships with the Discovery & Coverage app, per its own footer) — so before hand-building saturation tiles, open or import that dashboard: it already covers host vitals, the Java process, networking, and REST API health, with per-zone/group/instance filtering.

A live-tenant enumeration (July 2026) shows ~53 keys in the namespace. By area:

Area	Metrics	Notable dimensions / use
Host vitals	<code>system.cpu_usage</code> , <code>system.total_memory</code> / <code>system.free_memory</code> , <code>storage.volume.free</code> / <code>.total</code>	<code>volume</code>
Java process	<code>jvm.cpu_usage</code> , <code>jvm.heap_memory_used</code> / <code>_available</code> , <code>jvm.gc.major_collection_time</code> , <code>thread_pool.busy_threads</code> / <code>thread_pool.queue_size</code>	<code>thread_pool_name</code>
Networking	<code>communication.messages.dropped</code> / <code>.rejected</code> / <code>.resent</code> , <code>communication.agent_modules.connected</code> , <code>traffic.server.sent</code> / <code>.received</code> (clients ↔ AG), <code>traffic.client.sent</code> / <code>.received</code> (AG ↔ Dynatrace)	zone / group / instance
Event ingest	<code>event_ingest.event_incoming_count</code> , <code>.drop_count</code> , <code>.event_otlp_size</code> / <code>.event_json_size</code> , attribute counters	<code>operation</code> , <code>drop_reason</code> — <code>incoming_count</code> is the denominator that turns drops into a drop rate
Metrics ingest	<code>metrics.ingest.otlp.datapoints.received.total</code> vs <code>.accepted</code> , <code>metrics.ingest.grail.datapoints.accepted</code>	received minus accepted = OTLP metric datapoints rejected on the way in
REST API	<code>rest.request_count</code> , <code>rest.request_size</code> / <code>response_size</code> , <code>rest.response_time</code>	<code>operation</code> , <code>response_code</code>
Kubernetes module	<code>kubernetes.pipeline_duration</code> , <code>kubernetes.api.query_count</code> / <code>.query_duration</code> / <code>.connections.pool.available</code> , <code>kubernetes.events.observed</code> / <code>.processed</code> , cache size/evictions	<code>kubernetes.pipeline_duration</code> is the observable behind §4's "pipeline execution time > 50–60 s" scale-up threshold. <code>kubernetes.api.query_count</code> steps down at the 1.343 upgrade — see the note below.

Area	Metrics	Notable dimensions / use
AWS module	aws.data_delay , aws.data_lost_monitoring_status , aws.elements.bad / .reported / .total , aws.empty_responses , aws.query_time , aws.requests	health of the cloud-API-polling capability (§2) — delays/bad elements mean the polling AG is behind, not that AWS is down
Other modules	rum.beacon_forwarded_count , storage.directory.size / .limit (quotas by module_name), filecache.size_limit	

Rolling out (ActiveGate 1.343) — `kubernetes.api.query_count` will step down, and that is the improvement. ActiveGate 1.343 (published 07/15/2026, rollout from 07/28/2026) changes the Kubernetes module to call the Kubernetes API **once per minute**, down from multiple calls per collection cycle. So at the upgrade `kubernetes.api.query_count` drops sharply — this is the efficiency change landing, **not** the ActiveGate going quiet or the module failing.

The trap is in the direction of the reading. Falling query counts otherwise look like a module that has stopped collecting, and a rule written to alert on a *drop* will fire on the upgrade. Conversely, a rule reading the post-upgrade level as headroom is comparing against a baseline that no longer exists. **Re-baseline `kubernetes.api.query_count` after the upgrade reaches the cluster before treating its trend as a saturation signal at all** — and correlate against

`kubernetes.events.observed / .processed` and `kubernetes.pipeline_duration`, which reflect whether work is actually still being done. Verify the ActiveGate version in the cluster to know which side of the change a series is on.

A null zone/group dimension means the default zone/group — coalesce with `if(isNull(...), "default", else: ...)` before filtering or charting. Enumerate what your own tenant exposes — the catalog grows with the capabilities you run:

```
// Discover the ActiveGate self-monitoring catalog on your tenant
metrics
| filter startsWith(metric.key, "dt.sfm.active_gate")
| summarize series = count(), by:{metric.key}
| sort metric.key asc
```

One behavior to know: **drop/error metrics are sparse** — a key like `event_ingest.drop_count` only reports (and only enumerates) when drops actually occur. Chart them with `default: 0` (as the diagnostic dashboard does) so "no data" renders as the zero it means, and don't read a key's absence from the discovery query as it not existing.

The dashboard's own coloring thresholds are a useful reference point — slightly more conservative than, and consistent with, the 50%/80% machine guideline:

Signal	Amber at	Red at
System / JVM CPU, memory, heap	≥ 40%	≥ 70%
Storage volumes, directory quotas	≥ 50%	≥ 80%
JVM major-GC time (share of a 10-min interval)	≥ 5%	≥ 10%

Two signals from that catalog are worth watching beyond the four core signals, because they catch saturation the CPU/heap averages hide:

```
// GC pressure – major-collection time as a share of each 10-min interval
// The diagnostic dashboard colors this amber at 5%, red at 10%
timeseries gc_time = avg(dt.sfm.active_gate.jvm.gc.major_collection_time, rollup:sum),
from:-24h, interval:10m, by:{dt.active_gate.id, host.name}
| fieldsAdd gc_pct = 100 * duration(toLong(gc_time[]), "ms") / interval
| fieldsAdd peak_gc_pct = arrayMax(gc_pct)
| filter peak_gc_pct > 5
| sort peak_gc_pct desc
```

```
// Thread-pool backpressure – a growing queue means a pool can't keep up
timeseries queue_size = max(dt.sfm.active_gate.thread_pool.queue_size), from:-24h, by:
{dt.active_gate.id, host.name, thread_pool_name}
| fieldsAdd peak_queue = arrayMax(queue_size)
| filter peak_queue > 0
| sort peak_queue desc
```

And for the §6 HA math, a one-query fleet inventory — how many ActiveGates each zone actually has:

```
// ActiveGates per network zone – the denominator for the survivor-capacity check
timeseries cpu_samples = count(dt.sfm.active_gate.system.cpu_usage), from:-24h, by:
{dt.network_zone.id, dt.active_gate.id}
| fieldsAdd network_zone = if(isNull(dt.network_zone.id), "default", else:
dt.network_zone.id)
| summarize total_active_gates = countDistinctExact(dt.active_gate.id), by:{network_zone}
```

Pinpointing ingest loss by endpoint and reason

The namespace's sharpest diagnostic — and one the ready-made dashboard does *not* chart — turns "the ActiveGate dropped something" into "*this* endpoint dropped *this many* events for *this* reason." The load-bearing metric is `dt.sfm.active_gate.event_ingest.drop_count`, dimensioned by:

Dimension	What it splits by	Why it matters
operation	The ingest endpoint, e.g. <code>POST /otlp/v1/logs</code>	Isolates OTel log loss from other ingest paths

Dimension	What it splits by	Why it matters
<code>drop_reason</code>	<code>head_on_queue_limit</code> (in-memory ingest queue full — events dropped at the head) vs <code>disk_queue_limit</code> (disk buffer exhausted)	Tells you <i>which</i> buffer saturated, which changes the fix
<code>dt.active_gate.id</code> , <code>host.name</code> , <code>dt.active_gate.group.name</code> , <code>dt.network_zone.id</code> , <code>dt.active_gate.working_mode</code>	Fleet slicing	Pinpoints the ActiveGate and zone

```
// OTEL log loss at the ActiveGate, by machine and drop reason (24h)
timeseries drop_count = sum(dt.sfm.active_gate.event_ingest.drop_count), from:-24h, by:
{dt.active_gate.id, host.name, operation, drop_reason}, filter:{dt.system.bucket ==
"dt_system_metrics"}
| filter operation == "POST /otlp/v1/logs"
| fieldsAdd total_dropped = arraySum(drop_count)
| filter total_dropped > 0
| sort total_dropped desc
```

```
// Average OTLP event payload size per ActiveGate and signal type (24h)
// Rising payload size consumes routing capacity faster than the host-count tables assume
($3)
timeseries otlp_bytes = avg(dt.sfm.active_gate.event_ingest.event_otlp_size), from:-24h,
by:{dt.active_gate.id, host.name, event.type}, filter:{dt.system.bucket ==
"dt_system_metrics"}
| fieldsAdd avg_event_bytes = round(arrayAvg(otlp_bytes), decimals: 0)
| sort avg_event_bytes desc
```

Reading the `drop_reason` split: `head_on_queue_limit` is a throughput problem — the ActiveGate is receiving faster than it can process, so scale up/out or split capabilities (§8). `disk_queue_limit` means the disk buffer that absorbs uplink outages filled up — check connectivity to Dynatrace first, then the buffer allowance (the OPLOGS series documents the ActiveGate disk-queue behavior and its default size cap for log delivery). A companion metric, `dt.sfm.active_gate.communication.queue.outgoing.usage`, gives the outgoing-queue signal already used in the core queries above.

To judge severity rather than raw counts, pair `drop_count` against `event_ingest.event_incoming_count` on the same dimensions — a thousand drops out of a thousand incoming events is an outage, out of a billion it's noise.

The classic (Gen2) counterpart — `dsfm:`

The `dsfm:active_gate.*` family is the **Metrics Classic** form of the same self-monitoring signals — its reference page sits in the Metrics Classic docs section, and it's built for the Gen2 surfaces: metric selectors, Data Explorer, and the classic Metrics API. Two reasons it still matters even though this section queries Gen3 throughout:

- **It's the documented one.** The ActiveGate self-monitoring docs pages list `dsfm:` keys only — the `dt.sfm.*` keys used above appear on no docs page yet; their provenance is the shipped diagnostic dashboard plus live-tenant verification (including a full key enumeration), so confirm keys and dimension values in your own tenant before wiring alerts to them. The docs' authoritative signal interpretations — for the queue, *"growing value indicates messages received faster than sent out"*; for drops, *"non-zero value may indicate data loss"* — apply to both forms.
- **Classic surfaces still consume it.** Anything driven by a metric selector rather than DQL reads the `dsfm:` form — and on **Dynatrace Managed**, where Grail isn't available, `dsfm:` is the self-monitoring surface, full stop.

If you do query `dsfm:` keys in DQL, mind the colon: `timeseries` parses an unquoted `dsfm:` prefix as a parameter name, so backtick-quote the key — `avg(`dsfm:active_gate.jvm.cpu_usage`)`.

Sources: [ActiveGate self-monitoring metrics \(DT docs\)](#) — `dsfm:` metric keys, descriptions, and the queued/dropped interpretations quoted; [Self-monitoring metrics \(DT docs\)](#) — the `dsfm:` family's placement under Metrics Classic and its selector/Data-Explorer query surface; [Discovery & Coverage \(DT docs\)](#) — the app that provides the ActiveGate diagnostic overview dashboard; [Linux ActiveGate hardware and system requirements \(DT docs\)](#) — 50%/80% thresholds. [ActiveGate 1.343 release notes \(DT docs\)](#) — the Kubernetes module now calls the Kubernetes API once per minute, down from multiple calls per collection cycle. **Derived:** the Gen2/Gen3 labeling reads the docs' Metrics-Classic placement of `dsfm:` against `dt.sfm.*`'s Grail-bucket residency — Dynatrace doesn't state the generation split explicitly; mapping the 50%/80% machine guideline onto the JVM-level metrics as alert thresholds; the `dt.sfm.active_gate.*` catalog is grounded in a live-tenant key enumeration (July 2026, ~53 keys) plus the shipped diagnostic dashboard's definition (threshold colorings, `drop_reason` / `operation` dimension values) — not yet documented at the `sfm` page; all queries syntax-validated against a live tenant's DQL verifier.

8. Scale Up or Scale Out?

When §7 says an ActiveGate is running hot, the right move depends on which capability is saturating:

Signal	Capability	Move
CPU peaks past 50% on a routing ActiveGate	Routing	Scale out — add an ActiveGate to the zone; OneAgents distribute across the zone's ActiveGates, and the addition also buys failover headroom (§6)
Kubernetes-monitoring pod hits the §4 thresholds (working set >80%, pipeline >50–60 s)	K8s platform monitoring	Scale up — move to the next scenario's requests/limits; this workload <i>"scales vertically only"</i>
In-cluster routing replicas saturating	K8s OneAgent routing	Scale out — raise replicas per the §4 routing table

Signal	Capability	Move
Browser-monitor queue delays / executions near the node ceiling	Synthetic	Scale up the node (XS → S → M → L) or add nodes to the location — the location load-balances across nodes
Outgoing queue grows while the network is healthy	Any	Scale up CPU and memory first; if one ActiveGate carries several jobs, split the capabilities onto dedicated ActiveGates (§2)
One ActiveGate accumulating jobs over time	Mixed	Split before you supersize — a purpose-per-ActiveGate fleet is easier to size, update (FAQ-05), and troubleshoot

The general shape: **routing scales out, monitoring workloads scale up, synthetic scales by nodes** — and capability separation is the escape hatch when one box is doing too many different things for either axis to fix.

Sources: [Sizing guide for ActiveGates in the Kubernetes monitoring use-case \(DT docs\)](#) — vertical-only vs replicas split; [Private Synthetic locations \(DT docs\)](#) — node failover/load-balancing; [Get started with network zones \(DT docs\)](#) — zone distribution. **Derived:** the routing-scales-out generalization composes the zone failover model with the per-capability guidance; the split-before-supersize recommendation is community practice.

9. Recommended Approach

1. **Inventory the jobs first (§2).** List every capability each planned or existing ActiveGate carries: routing, Kubernetes monitoring, extensions, cloud API polling, synthetic, z/OS. If one ActiveGate carries more than routing plus light extensions, plan a split.
2. **Quantify each job's driver.** Hosts per zone for routing; pods per cluster for Kubernetes; browser executions/hour for synthetic; extension instances for EEC.
3. **Pick baselines from the current tables (§3–§5)** — and re-check the docs pages when you do, because the reference instances and estimates move between versions.
4. **Apply the HA multiplier (§6).** Per zone: enough ActiveGates that the fleet holds ≤50% CPU with one member down. Per synthetic location: at least two nodes.
5. **Deploy per the owning series** — ONBRD-03 for host-based install/groups/zones, K8S for DynaKube, SYNTH-04 for locations.
6. **Baseline the self-monitoring metrics on day one (§7)** and alert on CPU >50%, heap >80%, queue growth, and any dropped messages.
7. **Scale on signal, not on schedule (§8):** out for routing, up for Kubernetes monitoring, nodes for synthetic — and split capabilities when one ActiveGate does too much.
8. **Re-review at growth events:** onboarding waves, new clusters, monitor-schedule expansions, and before each migration phase (M2S-03/S2S-04 both put ActiveGate re-sizing in the design step for a reason).

10. Common Gotchas

Symptom	Likely cause	Where
Data gaps during an ActiveGate outage	Zone sized for average load, not survivor load — the remaining ActiveGates saturated	§6
<code>messages.dropped</code> non-zero but CPU looks fine	Outgoing queue backed up — uplink/proxy bottleneck, not machine size	§7
OTel logs going missing but the fleet-level dropped-message metrics look clean	Endpoint-level ingest drops — check <code>dt.sfm.active_gate.event_ingest.drop_count</code> by <code>operation</code> and <code>drop_reason</code> (<code>head_on_queue_limit</code> = throughput, <code>disk_queue_limit</code> = uplink/buffer)	§7
Processing lags but JVM CPU looks moderate	GC pressure — major-collection time above ~10% of the interval means the heap is too small for the workload	§7
Browser monitors intermittently delayed	Synthetic node at its executions/hour ceiling, or location has a single node	§5
Kubernetes-monitoring data lags on a big cluster	ActiveGate CPU-throttled by its limit, or pipeline time past 50–60 s	§4
One ActiveGate monitoring several clusters misbehaves	Multi-cluster-per-ActiveGate is documented as not recommended	§4
Extensions stop persisting data	EEC lacks its required ~2.1 GB free-disk allowance	§3
"The table said 800 hosts but we saturated at 500"	Estimates assume typical per-host volume — heavy log routing or dense instrumentation consumes it faster	§3, §7
Everything degraded after adding one more job to a shared ActiveGate	Capability stacking — demanding jobs deserve dedicated ActiveGates	§2, §8
Zone failure overloaded a <i>different</i> zone's ActiveGates	Fallback mode <i>Any</i> ActiveGate rerouted the load somewhere unsized for it	§6

Sources: each row maps to the documented constraint or threshold cited in the referenced section — see the Sources blocks of §3–§8.

 **Disclaimer:** This content is AI-generated, community-driven, and **not supported by Dynatrace**. ActiveGate sizing tables, supported platforms, and self-monitoring metrics evolve across releases — always verify current numbers against the [Dynatrace documentation](#) and your own tenant's self-monitoring metrics before committing hardware.